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Electronic control lock in combination with kickstand of a bicycle

Abstract

An electronic control lock of a bicycle has an electronic controller, a kickstand lock, and a frame lock. The kickstand lock includes a kickstand and a stand control device. When locking, the stand control device stops the kickstand from being kicked up, causing inconvenience or impossible to ride the bicycle. The frame lock includes an inserting element and a frame control device. When locking, the frame control device makes the inserting element unable to be pulled out. Thus the inserting element can fasten the bicycle with a chain. When unlocking, the stand control device and the frame control device are controlled by the electronic controller, which is controlled by a user, to allow the kickstand being kicked up and the inserting element being pulled out. The kickstand lock and the frame lock can be locked or unlocked without traditional keys.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a lock of a bicycle, especially to a frame lock and a kickstand lock that can be electronically controlled, and a locking/unlocking method of the frame lock and the kickstand lock.

2. Description of the Prior Art(s)

(2) In modern society, in order to prevent a bicycle from being stolen, users often buy locks to lock the bicycle at a specification position or to prevent the wheels of the bicycle from turning. However, a conventional lock for the bicycles is locked or unlocked by inserting a key, which is inconvenient to use. Therefore, the conventional lock for the bicycles needs to be improved.

(3) To overcome the shortcomings, the present invention provides an electronic control lock of a bicycle and an electronically controlled locking/unlocking method of a bicycle to mitigate or obviate the aforementioned problems.

SUMMARY OF THE INVENTION

(4) The main objective of the present invention is to provide an electronic control lock of a bicycle. The electronic control lock comprises an electronic controller and a kickstand lock. The kickstand lock includes a kickstand and a stand control device. The kickstand is rotatable and has a flip-down state and a kicked-up state. The stand control device is in signal connection with the electronic controller, is controlled by the electronic controller, and has a stand locking state and a stand unlocking state. In the stand locking state, the stand control device stops the kickstand that is in the flip-down state from being rotated to the kicked-up state. In the stand unlocking state, the kickstand is rotatable to the flip-down state and the kicked-up state.

(5) The main objective of the present invention is to provide an electronic control lock of a bicycle. The electronic control lock comprises an electronic controller and a frame lock. The frame lock includes a housing, an inserting element, and a frame control device. The housing has an insertion hole. The inserting element is detachably inserted in the insertion hole. The frame control device is in signal connection with the electronic controller, is controlled by the electronic controller, and has a frame locking state and a frame unlocking state. In the frame locking state, the frame control device stops the inserting element from detaching from the insertion hole. In the frame unlocking state, the inserting element is detachable from the insertion hole.

(6) The main objective of the present invention is to provide an electronically controlled locking/unlocking method of a bicycle. When a kickstand has been flipped down, a stand control device makes the kickstand unable to be kicked up. When unlocking the kickstand, the stand control device is controlled by sending signals through an electronic controller, thereby the kickstand is able to be kicked up.

(7) The main objective of the present invention is to provide an electronically controlled locking/unlocking method of a bicycle. When an inserting element has been inserted into an insertion hole, a frame control device makes the inserting element unable to detach from the insertion hole. When unlocking the inserting element, the frame control device is controlled by sending signals through an electronic controller, thereby the inserting element is detachable from the insertion hole.

(8) In the present invention, the stand control device controls rotation of the kickstand, and the frame control device controls whether the inserting element can be pulled out. When the kickstand is locked, the stand control device stops the kickstand that is in the flip-down state from being kicked up. Thus, the kickstand keeps flipping down, which causes inconvenience to ride the bicycle or even stops a pedal from rotating. When the frame lock is locked, the frame control device stops the inserting element from being pulled out of the insertion hole. The inserting element can be connected with a chain, so as to fasten the bicycle to a specific place. Besides, the chain may be mounted through a wheel of the bicycle, so as to prevent the wheel from rotating.

(9) When unlocking, a user may control the electronic controller in various ways, such as through an application in a mobile phone. The stand control device is controlled by the electronic controller to allow the kickstand being kicked up, making it easy to ride the bicycle. The frame control device is controlled by the electronic controller to allow the inserting element being pulled out or be ejected automatically. Therefore, the kickstand lock and the frame lock can be unlocked without traditional keys.
(10) In addition, since the kickstand is controlled by the stand control device and the inserting element is controlled by the frame control device, the kickstand and the inserting element may be designed with automatic locking function. For instance, when the kickstand is kicked to flip down, the stand rotating device is automatically switched to a locking state, or when the inserting element has been inserted into the insertion hole, the frame control device is automatically switched to a locking state. Thus, the kickstand lock and the frame lock can be locked without traditional keys.
(11) Other objectives, advantages and novel features of the invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1** and **2** are operational side views of a first embodiment of an electronic control lock in accordance with the present invention, showing mounted on a bicycle;
(2) FIG. **3** is an exploded perspective view of an electronic controller of the first embodiment of the electronic control lock in FIG. **1**;
(3) FIG. **4** is a perspective view of a kickstand lock of the first embodiment of the electronic control lock in FIG. **1**;
(4) FIGS. **5** and **6** are exploded perspective views of the kickstand lock of the first embodiment of the electronic control lock in FIG. **4**;
(5) FIGS. **7** to **10** are operational side views of the kickstand lock of the first embodiment of the electronic control lock in FIG. **4**;
(6) FIG. **11** is an exploded perspective view of a frame lock of the first embodiment of the electronic control lock in FIG. **1**;
(7) FIGS. **12** and **13** are exploded perspective views of the frame lock of the first embodiment of the electronic control lock in FIG. **11**;
(8) FIGS. **14** to **17** are operational side views in partial section of the frame lock of the first embodiment of the electronic control lock in FIG. **11**;
(9) FIGS. **18** and **19** are enlarged perspective views of another embodiment of a frame lock in accordance with the present invention;
(10) FIG. **20** is an enlarged perspective view of a second embodiment of an electronic control lock in accordance with the present invention;
(11) FIG. **21** is an enlarged side view of the second embodiment of an electronic control lock in FIG. **20**; and
(12) FIG. **22** is an operational perspective view of another embodiment of a kickstand in accordance with the present invention, showing mounted on a bicycle.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(13) With reference to FIGS. **1** and **2**, a first embodiment of an electronic control lock of a bicycle in accordance with the present invention comprises, but not limited to, an electronic controller **10**, a kickstand lock **20**, and a frame lock **30**. In other embodiments, the electronic control lock of the bicycle may only comprise the electronic controller **10** and the kickstand lock **20**, or only comprise the electronic controller **10** and the frame lock **30**.

(14) With reference to FIGS. **1** to **3**, the electronic controller **10** may be mounted anywhere on the bicycle, as long as the electronic controller **10** is able to be electrically connected to the kickstand lock **20** and the frame lock **30**. In addition, the electronic controller **10** may be electrically connected to the kickstand lock **20** and the frame lock **30** in a wired or wireless manner. Preferably, the electronic controller **10** has a Controller Area Network Bus (CAN bus) system and a Bluetooth Low Energy (BLE) system and may also be equipped with a Global Positioning System (GPS).
(15) With reference to FIGS. **1** and **2**, the kickstand lock **20** may also be mounted anywhere on the bicycle, as long as a kickstand **21** can be propped on the ground when being flipped down. Preferably, as shown in FIG. **1**, the kickstand lock **20** may be mounted beside a pedal **40**, so as to prevent the pedal **40** from turning. Besides, as shown in FIG. **2**, the kickstand lock **20** may be mounted to an axle of a rear wheel **60**, which is a common position for mounting the kickstand **21**.
(16) With reference to FIGS. **1** and **2**, the frame lock **30** may also be mounted anywhere on the bicycle. Preferably, as shown in FIG. **1**, the frame lock **30** may be mounted beside a front wheel **50** or the rear wheel **60**, so as to directly lock the front wheel **50** or the rear wheel **60** (such as with a chain **321**). However, it is not limited thereto, and the frame lock **30** may also be fastened to a parking rack. In this way, the frame lock **30** does not have to be disposed adjacent to the front wheel **50** or the rear wheel **60**.
(17) With reference to FIG. **4**, the kickstand lock **20** includes a kickstand **21** and a stand control device **22**.
(18) With reference to FIGS. **1** and **4**, the kickstand **21** is rotatable and has a flip-down state and a kicked-up state. When the kickstand lock **20** is mounted beside the pedal **40**, it is the kickstand **21** that is disposed beside the pedal **40**. As the kickstand **21** is in the flip-down state, the kickstand **21** prevents the pedal of the bicycle from turning, so as to provide anti-theft effect. As for the kickstand **21** that prevents the pedal **40** from turning, it is not limited to make the pedal **40** be stuck by the kickstand **21** completely and may be just prevent the pedal **40** from turning (interfere with the turning of the pedal **40**).
(19) The stand control device **22** is in signal connection with the electronic controller **10** (wired or wireless), is controlled by the electronic controller **10**, and has a stand locking state and a stand unlocking state. In the stand locking state, the stand control device **22** stops the kickstand **21** that is in the flip-down state from being rotated to the kicked-up state. In the stand unlocking state, the kickstand **21** is rotatable to the flip-down state and the kicked-up state.
(20) With reference to FIGS. **5** to **7**, a structure, which stops the kickstand **21** from being kicked up, of the stand control device **22** is shown. Preferably, the stand control device **22** includes a stand engaging element **221**, a stand resilient element **222**, and a stand operating element **223**. The stand resilient element **222** tends to force the stand engaging element **221** to engage with the kickstand **21**, so as to stop the kickstand **21** from rotating. The stand resilient element **222** is preferably a compression spring. The stand operating element **223** is in signal connection with the electronic controller **10** and is controlled by the electronic controller **10**. With reference to FIG. **7**, in the stand locking state, the stand operating element **223** stops the stand engaging element **221** from disengaging with the kickstand **21**. With reference to FIGS. **8** and **9**, in the stand unlocking state, the stand operating element **223** allows the stand engaging element **221** to disengage with the kickstand **21**.
(21) With reference to FIGS. **6**, **7** and **9**, an engaging structure between the stand engaging element **221** and the kickstand **21** is shown. Preferably, a first engaging portion **2211** is formed on the stand engaging element **221**, and a second engaging portion **211** is formed on the kickstand **21** and corresponds in shape to the first engaging portion **2211**. When in the stand locking state and the stand unlocking state, the second engaging portion **211** engages with the first engaging portion **2211**, so as to stop the kickstand **21** from rotating. When the kickstand **21** rotates, the second engaging portion **211** and the first engaging portion **2211** drives the stand engaging element **221** to move away from the kickstand **21**. Preferably, the first engaging portion **2211** includes multiple

triangular recesses and multiple triangular protrusions, and the second engaging portion **211** also includes multiple triangular recesses and multiple triangular protrusions. However, shapes of the first and second engaging portions **2211**, **211** are not limited thereto.
(22) A structure, which stops the stand engaging element **221** from disengaging from the stand operating element **223** is as described as follows. Preferably, the stand operating element **223** includes a stand rotating device **2231** and a stand rotating seat **2232**. The stand rotating device **2231** is in signal connection with the electronic controller **10** and is controlled by the electronic controller **10** to rotate the stand rotating seat **2232**. Preferably, the stand rotating device **2231** is, but is not limited to, a motor. At least one stand protrusion **2233** and at least one stand recess **2234** are formed on the stand rotating seat **2232** and are arranged annularly.
(23) With reference to FIG. **7**, when in the stand locking state, the stand rotating device **2231** rotates the stand rotating seat **2232** to make the at least one stand protrusion **2233** abut against the stand engaging element **221**, so as to stop the stand engaging element **221** from disengaging from kickstand **21**.
(24) With reference to FIGS. **8** and **9**, when in the stand unlocking state, the stand rotating device **2231** rotates the stand rotating seat **2232** to make the at least one stand recess **2234** corresponds in position to the stand engaging element **221**, so as to allow the stand engaging element **221** to move toward the at least one stand recess **2234** and to disengage from the kickstand **21**.
(25) With reference to FIGS. **6**, **8** and **9**, the way that the at least one stand protrusion **2233** and the at least one stand recess **2234** cooperate with the stand engaging element **221** is shown. Preferably, an acting protrusion **2212** protrudes from the stand engaging element **221** and corresponds in position to the at least one stand protrusion **2233** and the at least one stand recess **2234**. In the first embodiment, the stand rotating seat **2232** is disposed on a diametrical side of the stand engaging element **221**, such that the acting protrusion **2212** radially protrudes from the stand engaging element **221**. However, it is not limited thereto. The stand rotating seat **2232** also may be disposed on an axial side of the stand engaging element **221**, i.e. the stand engaging element **221** is disposed between the stand rotating seat **2232** and the kickstand **21**. Thus, the acting protrusion **2212** axially protrudes toward the stand rotating seat **2232** and engages with the stand engaging element **221**.
(26) In addition, the stand operating element **223** is not limited to the structure that rotates. The stand operating element **223** also may be a structure that moves linearly. For instance, the stand operating element **223** may be a pressure cylinder or a linear motor. When locking, the stand operating element **223** moves linearly (such as protruding) to abut against the stand engaging element **221**, so as to stop the stand engaging element **221** from disengaging from kickstand **21**. When unlocking, the stand operating element **223** moves linearly in reverse direction (such retracting), so as to allow the stand engaging element **221** to move and to disengage from the kickstand **21**.
(27) In summary, with reference to FIG. **7**, when locking the kickstand **21**, the stand rotating device **2231** rotates the stand rotating seat **2232** to a specific angle to make the stand engaging element **221** unable to move away from kickstand **21**. Accordingly, the kickstand **21** is unable to be kicked up. Thus, the kickstand **21** keeps flipping down, which causes inconvenience to ride the bicycle or stops the pedal **40** from rotating. With reference to FIGS. **8** to **10**, when unlocking the kickstand **21**, the stand rotating seat **2232** is rotated to allow the stand engaging element **221** to move and to disengage from the kickstand **21**. Thus, the kickstand **21** can be kicked up.
(28) In addition, the stand control device **22** may be designed with automatic locking function. That is, when the kickstand **21** is rotated to flip down, the stand rotating device **2231** would automatically rotate the stand rotating seat **2232** to said specific angle to make the stand engaging element **221** unable to move away from kickstand **21**.
(29) Moreover, a sensing device may be additionally mounted to the kickstand lock **20** to identify a rotating angle of the kickstand **21** and to use with the aforementioned application scenarios.
(30) In addition, with reference to FIG. **4**, the kickstand **21** may be a single elongated rod, such that

the bicycle tilts toward the kickstand **21**. With reference to FIG. **22**, the kickstand **21** may be formed to have two standing rods **212** and a bottom rod **213**, such that the rear wheel **60** can be lifted up.

(31) With reference to FIG. **11**, the frame lock **30** includes a housing **31**, an inserting element **32**, and a frame control device **33**.

(32) The housing **31** has an insertion hole **311**.

(33) The inserting element **32** is detachably inserted in the insertion hole **311** of the housing **31**. As shown in FIG. **1**, an end of the inserting element **32** may be connected with other devices, such as the chain **321**, that are used to be fastened to other places.

(34) The frame control device **33** is in signal connection with the electronic controller **10** (wired or wireless), is controlled by the electronic controller **10**, and has a frame locking state and a frame unlocking state. In the frame locking state, the frame control device **33** stops the inserting element **32** from detaching from the insertion hole **311**. In the frame unlocking state, the inserting element **32** is detachable from the insertion hole **311**.

(35) With reference to FIGS. **12** to **14**, a structure, which stops the inserting element **32** from detaching from the insertion hole **311**, of the frame control device **33** is shown. Preferably, the frame control device **33** includes a frame engaging element **331**, a frame resilient element **332**, and a frame operating element **333**. The frame resilient element **332** tends to force the frame engaging element **331** to engage with the inserting element **32**, so as to stop the inserting element **32** from detaching from the insertion hole **311**. The frame resilient element **332** is preferably a micro switch that provides sufficient resilient force to push the frame engaging element **331**. The frame resilient element **332**, which is micro switch, is also able to identify a position of the frame engaging element **331**. Besides, the frame engaging element **331** may also be a compression spring. The frame operating element **333** is in signal connection with the electronic controller **10** and is controlled by the electronic controller **10**. With reference to FIG. **17**, in the frame unlocking state, the frame operating element **333** makes the frame engaging element **331** disengage from the inserting element **32**, such that the inserting element **32** is detachable from the insertion hole **311**.

(36) With reference to FIGS. **13**, **16** and **17**, a structure, which makes the frame engaging element **331** disengage from the inserting element **32**, of the frame operating element **333** is shown. Preferably, the frame operating element **333** includes a frame rotating device **3331** and a frame rotating seat **3332**. The frame rotating device **3331** is in signal connection with the electronic controller **10** and is controlled by the electronic controller **10** to rotate the frame rotating seat **3332**. Preferably, the frame rotating device **3331** is, but is not limited to, a motor. At least one frame protrusion **3333** is formed on the frame rotating seat **3332**. Preferably, the frame protrusion **3333** can be a bulging part on a cam.

(37) With reference to FIG. **16**, when in the frame locking state, the frame resilient element **332** makes the frame engaging element **331** engage with the inserting element **32**.

(38) With reference to FIG. **17**, when in the frame unlocking state, the frame rotating device **3331** drives the frame rotating seat **3332** to rotate to make the at least one frame protrusion **3333** abut against the frame engaging element **331**, so as to make the frame engaging element **331** disengage from the inserting element **32**. Thus the inserting element **32** is detachable from the insertion hole **311**.

(39) In addition, the aforementioned structure may be designed as a free latching tongue. With reference to FIGS. **14** to **16**, for instance, when inserting the inserting element **32** into the insertion hole **311**, the inserting element **32** pushes the frame engaging element **331** to move the frame engaging element **331** by compressing the frame resilient element **332**. When the inserting element **32** has been inserted into the insertion hole **311** to a certain depth, the frame resilient element **332** pushes the frame engaging element **331** to engage with the inserting element **32**, so as to stop the inserting element **32** from detaching the insertion hole **311**.

(40) With reference to FIGS. **13** to **16**, the frame engaging element **331** is preferably an annular

component in one embodiment. Thus, when the inserting element **32** is inserted into the insertion hole **311**, the inserting element **32** is mounted through the frame engaging element **331** and abuts against a circumference of the frame engaging element **331** to move the frame engaging element **331** to compress the frame resilient element **332**. Preferably, an annular inclined surface **322** is formed on an end of the inserting element **32** and an inclined abutting surface **3311** is formed on the circumference of the frame engaging element **331**. The annular inclined surface **322** and the inclined abutting surface **3311** facilitate the inserting element **32** to push the frame engaging element **331** in a radial direction. However, it is also possible to have only the annular inclined surface **322** or only the inclined abutting surface **3311**.
(41) With reference to FIGS. **18** and **19**, the frame engaging element **331** is preferably an elongated component in another embodiment and has an inner pivotally connected to elsewhere (such as the frame operating element **333**). The frame resilient element **332** pushes the frame engaging element **331** to rotate, so as to make an outer end of the frame engaging element **331** engage with the inserting element **32**. When unlocking, the frame rotating seat **3332** is rotated and the at least one frame protrusion **3333** pushes the frame engaging element **331** to rotate, so as to disengage the outer end of the frame engaging element **331** from the inserting element **32**.
(42) Lastly, with reference to FIGS. **13**, **16** and **17**, in one preferred embodiment, when unlocking, the frame control device **33** pushes the inserting element **32** toward and out of the insertion hole **311**, but it is not limited to push the whole inserting element **32** out of the insertion hole **311**. Preferably, the frame control device **33** includes an ejecting resilient element **334**. When the inserting element **32** is inserted into the insertion hole **311** and the frame engaging element **331** engages with the inserting element **32**, the ejecting resilient element **334** is compressed. Once unlocked, the frame engaging element **331** disengages from the inserting element **32** and the ejecting resilient element **334** immediately pushes the inserting element **32** outward.
(43) In addition, the frame operating element **333** is not limited to the structure that rotates. The frame operating element **333** also may be a structure that moves linearly. For instance, the frame operating element **333** may be a pressure cylinder or a linear motor. When unlocking, the frame operating element **333** moves linearly (such as protruding) to abut against the frame engaging element **331**, so as to disengage the frame engaging element **331** from the inserting element **32**. When locking, the frame operating element **333** moves linearly in reverse direction (such retracting), so as to allow the frame resilient element **332** to push the frame engaging element **331** to engage with the inserting element **32**.
(44) In summary, when locking, the frame rotating device **3331** rotates the frame rotating seat **3332** to a specific angular range and be free from abutting the frame engaging element **331**. Thus, the frame resilient element **332** pushes the frame engaging element **331** to engage with the inserting element **32**. When unlocking, the frame rotating seat **3332** rotates and abuts against the frame engaging element **331**, such that the frame engaging element **331** disengage from the inserting element **32** and the ejecting resilient element **334** pushes the inserting element **32** to move outward.
(45) In addition, a sensing device may be additionally mounted to the frame lock **30** to identify if the inserting element **32** is inserted into the insertion hole **311** and to use with the aforementioned application scenarios
(46) With the kickstand lock **20** and the frame lock **30** as described above, in the present invention, the stand control device **22** controls rotation of the kickstand **21**, the frame control device **33** controls whether the inserting element **32** can be pulled out, and the electronic controller **10** controls the stand control device **22** and the frame control device **33**. A user may control the electronic controller **10** through an application, which is in connection with the electronic controller **10**, in a mobile phone. However, the electronic controller **10** may also be controlled through other ways.
(47) Specifically, when unlocking, the electronic controller **10** controls the stand control device **22** and the frame control device **33** to allow the kickstand **21** to be kicked up and the inserting element

32 to be pulled out or be ejected automatically.
(48) When locking, the stand control device **22** locks the kickstand **21** automatically as the kickstand **21** is kicked to flip down, and the frame control device **33** locks the inserting element **32** automatically as the inserting element **32** is inserted into the insertion hole **311**.
(49) Therefore, the kickstand lock **20** and the frame lock **30** can be locked or unlocked without traditional keys.
(50) Furthermore, the electronic control lock of the present invention may further comprises an alarm device. If the kickstand **21** is forced to rotate when the kickstand **21** is locked, if the inserting element **32** is pulled outward when the frame lock **30** is locked, or when the bicycle is shaken to a certain extent, the alarm device would sound and/or the electronic controller **10** would inform the user (such as through the application of the mobile phone) immediately.
(51) With reference to FIGS. **20** and **21**, a second embodiment of the electronic control lock is shown and is similar to the first embodiment. The differences therebetween are as follows. The kickstand lock **20** and the frame lock **30** are combined. The kickstand **21** and the inserting element **32** share one control device, which may be regarded as the frame control device **33** or the stand control device. The stand engaging element **221** and the frame engaging element **331** share one operating element, which may be regarded as the frame operating element **333** or the stand operating element. That is, the stand engaging element **221** and the frame engaging element **331** share one rotating device, which may be regarded as the frame rotating device **3331** or the stand rotating device, and one rotating seat, which may be regarded as the frame rotating seat **3332** or the stand rotating seat. The at least one stand protrusion **2233**, the at least one stand recess **2234**, and the at least one frame protrusion **3333** are disposed on the rotating device at the same time. Preferably, the at least one frame protrusion **3333** is formed as a closed track, and the at least one stand protrusion **2233** and the at least one stand recess **2234** are formed as an open structure (facing downward). However, they are not limited thereto.
(52) In the second embodiment, rotation of the single rotating seat controls whether the kickstand **21** is able to be kicked up and whether the inserting element **32** is able to be detached. Therefore, the whole structure can be further simplified.
(53) An electronically controlled locking/unlocking method of the bicycle has four embodiments. The first embodiment has only the kickstand **21**, the second embodiment has only the frame lock **30**, and each of the third and fourth embodiments has both the kickstand **21** and the frame lock **30**.
(54) The first embodiment of the electronically controlled locking/unlocking method includes steps as follows.
(55) With reference to FIGS. **4** and **7**, when the kickstand **21** has been flipped down, the stand control device **22** makes the kickstand **21** unable to be kicked up. Preferably, when the kickstand **21** has been flipped down, the stand control device **22** automatically makes the kickstand **21** unable to be kicked up. However, it is not limited thereto. The stand control device **22** can be controlled by human to make the kickstand **21** unable to be kicked up.
(56) With reference to FIGS. **8** to **10**, when unlocking the kickstand **21**, the stand control device **22** is controlled by sending signals through the electronic controller **10**, such that the kickstand **21** is able to be kicked up.
(57) The second embodiment of the electronically controlled locking/unlocking method includes steps as follows.
(58) With reference to FIGS. **11** and **16**, when the inserting element **32** has been inserted into the insertion hole **311**, the frame control device **33** makes the inserting element **32** unable to detach from the insertion hole **311**. Preferably, when the inserting element **32** has been inserted into the insertion hole **311**, the frame control device **33** automatically makes the inserting element **32** unable to detach from the insertion hole **311**. However, it is not limited thereto.
(59) With reference to FIG. **17**, when unlocking the inserting element **32**, the frame control device **33** is controlled by sending signals through the electronic controller **10**, such that the inserting

element **32** is detachable from the insertion hole **311**.

(60) The third embodiment of the electronically controlled locking/unlocking method includes steps as follows.

(61) When the kickstand **21** has been flipped down, the stand control device **22** makes the kickstand **21** unable to be kicked up. When the inserting element **32** has been inserted into the insertion hole **311**, the frame control device **33** makes to the inserting element **32** unable to detach from the insertion hole **311**. Flipping down the kickstand **21** and inserting the inserting element **32** are performed in no particular order.

(62) When unlocking, the stand control device **22** is controlled by sending signals through the electronic controller **10** such that the kickstand **21** is able to be kicked up, and the frame control device **33** is controlled by sending signals through the electronic controller **10** such that the inserting element **32** is detachable from the insertion hole **311**. Unlocking the kickstand **21** and unlicking the inserting element **32** are performed in no particular order and may be performed simultaneously.

(63) The third embodiment of the electronically controlled locking/unlocking method is similar to the third embodiment. The differences therebetween are that the kickstand **21** and the inserting element **32** share one control device (may be regarded as the frame control device **33** or the stand control device). When the kickstand **21** has been flipped down, the control device makes the kickstand **21** unable to be kicked up. When the inserting element **32** has been inserted into the insertion hole **311**, said same control device makes to the inserting element **32** unable to detach from the insertion hole **311**.

(64) When unlocking, the stand control device **22** is controlled to allow the kickstand **21** being able to be kicked up and the frame control device **33** is controlled to allow the inserting element **32** being detachable from the insertion hole **311** by sending signals through the electronic controller **10**.

(65) The above-mentioned four embodiments of the electronically controlled locking/unlocking methods may be performed with the above-mentioned electronic control lock of the bicycle. Moreover, how the electronic control lock is used as described above can be incorporated into the electronically controlled locking/unlocking method. For instance, the inserting element **32** would be ejected and the alarm device would sound when the inserting element **32** is unlocked. More detailed descriptions are not repeated again.

(66) In addition, when controlling the electronic controller **10** in the above-mentioned four embodiments of the electronically controlled locking/unlocking methods, after commanding for each action, it can automatically check battery before starting the action. Moreover, before or after controlling the stand rotating device **2231** or the frame rotating device **3331** to rotate, it can automatically check whether the stand rotating device **2231** and the frame rotating device **3331** are disposed at or rotated to correct angles. In addition, when the kickstand **21** is kicked up or flipper down, it can also check whether the kickstand **21** is rotated to a correct angle. For instance, when unlocking and kicking up the kickstand **21**, if the kickstand **21** is not fully kicked up, the kickstand **21** would be automatically locked again. Furthermore, when the inserting element **32** has been inserted into or pulled out from the insertion hole **311**, it can automatically check whether the inserting element **32** is inserted to a correct position or is fully pulled out.

(67) Even though numerous characteristics and advantages of the present invention have been set forth in the foregoing description, together with details of the structure and features of the invention, the disclosure is illustrative only. Changes may be made in the details, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

1. An electronic control lock in combination with a kickstand of a bicycle, the combination comprising: an electronic controller; and a kickstand lock including the kickstand being rotatable and having a flip-down state and a kicked-up state; and a stand control device being in signal connection with the electronic controller, controlled by the electronic controller, and having a stand locking state and a stand unlocking state, wherein the stand control device includes a stand engaging element; a stand resilient element tending to force the stand engaging element to engage with the kickstand, so as to stop the kickstand from rotating; and a stand operating element being in signal connection with the electronic controller, controlled by the electronic controller, and including a stand rotating device and a stand rotating seat, wherein the stand rotating device is in signal connection with the electronic controller, is controlled by the electronic controller, and selectively rotates the stand rotating seat; and at least one stand protrusion and at least one stand recess are formed on the stand rotating seat and are arranged annularly; wherein in the stand locking state, the stand rotating device rotates the stand rotating seat to locate the at least one stand protrusion, such that when the kickstand is rotated out of the flip-down state, the stand engaging element directly abuts against the at least one stand protrusion, thereby stopping the kickstand from being further rotated to the kicked-up state; and wherein in the stand unlocking state, the stand rotating device rotates the stand rotating seat to make the at least one stand recess correspond in position to the stand engaging element, so as to allow the stand engaging element to move toward the at least one stand recess and to disengage from the kickstand, and thereby allowing the kickstand to rotate from the flip-down state to the kicked-up state.